

CS 4644/7643: Deep Learning Final Project Report & Rubric

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Discussions: <https://piazza.com/class/mk600jc9ifi304>

Due: 4 May 2026, 11:59 pm ET

Logistics

1. **One PDF per group.**
2. **Where/format.** Submit a PDF on Gradescope to **Final Report** (*Group Submission*; add all team members). Page limit: **6–8 pages** for main text (references and Team Contributions table do not count). Supplementary material is optional and not guaranteed to be reviewed. Because of the tight time period between the deadline and the end of the semester grading deadline, **there is no late due date. Submissions received after the deadline will receive a 0.**
3. **Templates.** This year we are using the ICLR conference template. You can just use the template from the milestone, or even just directly add to your milestone for the final report. You are welcome to reuse content from your milestone if appropriate. While creating the report, add all team members as authors.

Purpose of the Final Report

The goal of the final report is to clearly communicate your problem, approach, and findings. Strong reports justify design and experimental choices with evidence, analyze results (including failures), and demonstrate a deep understanding of how deep learning concepts apply to your project.

Your Final Report Should Include the Following Sections

1. Abstract (≤ 300 words)

Provide a compact, self-contained summary that a non-expert in your sub-area can understand. Include:

- the task, setting, and why it matters (1–2 sentences);
- your core idea in one clear sentence, and ideally what makes it different (e.g., loss, architecture, data regime);
- 1–2 headline numbers with metric, dataset/split, and relative/absolute gains.

2. Introduction

Convince the reader the problem is important and precisely defined. Include:

- what you tried to do, framed as a crisp research question and success criteria (*what constitutes success? which metric/split?*);
- who benefits (application domain/users), what gap is filled (scientific/engineering), and the expected implications if successful (e.g., lower compute, better generalization, safer deployment).

3. Problem Formulation + Related Work

Situate your work and clarify novelty without exhaustive surveying. Include:

- at least 3 closely related lines of work (baselines you will compare to, key prior ideas you build on); briefly note datasets commonly used;
- a comparison table or paragraph that makes your contributions explicit (assumptions, capabilities, trade-offs) and explains *how* you differ (e.g., data efficiency, robustness, complexity);
- mathematical description of the problem you are solving, specify inputs and desired outputs.

4. Method/Approach

Describe *what* you built and *why* it should work. Include:

- **CS7643 Groups Only:** 3 hypotheses (H1–H3), describing experiments, the results you expect to see, and why
- technical descriptions of training pipeline; architecture diagram or pseudocode; losses/objectives; regularizers; optimizer, LR schedule, batch size, epochs/steps; early stopping/selection rule;
- justification for design decisions;
- evaluation strategy (datasets/benchmarks tested on, evaluation metrics with definitions, and baselines);
- training framework, hardware, key hyperparameters (with defaults).

5. Data

Document the data so results are interpretable and replicable. Include:

- source; size; modalities; label space; class/attribute balance; known biases/limitations;
- train/val/test construction, normalization/filters, augmentation policy + justification.

6. Experiments and Results

Provide a fair, rigorous, and insightful empirical study. Include:

- 1–3 quantitative and/or qualitative baseline results (classical and/or prior SOTA);
- experiments + ablations; **CS7643 Groups Only:** confirm/refute H1–H3;
- your results in visually appealing plots/tables
- qualitative visualizations; discuss errors and potential fixes;

7. **Conclusion**

Summarize findings and future directions. Include:

- Discussion of results, what worked/failed + why, explicit limitations (data/compute/assumptions);
- concrete, feasible next steps or applications informed by your findings.

8. **Team Contributions** (required; not in page limit)

Provide a table with one row per member covering technical contributions (implementation, experiments, analysis, writing of technical sections). Administrative-only roles are insufficient.

9. **References** (required; not in page limit)

Include all cited works; ensure every reference is cited in-text and vice versa; use a consistent bibliographic style.

Grading Rubric for Groups *with* 7643 students (100 points)

Category & Criteria	Points
Abstract (5)	
Problem statement (1–2 sentences)	1
Summary of core approach	2
Headline results with metrics	2
Introduction (10)	
Visually-compelling and effective teaser figure	2
Objectives and crisp research question with success criteria	4
Impact: who benefits, what gap is filled, expected implications	4
Problem Formulation + Related Work (12)	
Coverage of at least 3 closely related lines of work	4
Positioning: comparison table/paragraph clarifying contributions	4
Description of problem with inputs/outputs and formulation	4
Method/Approach (24)	
3 testable hypotheses (H1–H3)	6
Description of model/algorithm + training details	6
Justification for design decisions	4
Evaluation strategy (datasets/benchmarks, metrics, baselines)	4
Reproducibility details (framework, hardware, hyperparameters)	4
Data (8)	
Description & datasheet aspects (source, size, balance, biases)	4
Description of preprocessing/augment steps + why and data splits	4
Experiments and Results (24)	
Strong, justified baselines (1–3)	6
Ablations & experiments	6
Confirmation/Refutation of hypotheses H1–H3	6
Visualizations/plots, analysis, and failure modes	6
Conclusion (5)	
Discussion of results and limitations	3
Concrete future work informed by findings	2
Additional (12)	
Clarity, formatting, template compliance (6–8 pages main text)	4
Team contribution table + descriptions of work	4
Proper references cited in consistent bibliographic style	4
Total	100

Grading Rubric for Groups *without* 7643 students (100 points)

Category & Criteria	Points
Abstract (5)	
Problem statement (1–2 sentences)	1
Summary of core approach	2
Headline results with metrics	2
Introduction (10)	
Visually-compelling and effective teaser figure	2
Objectives and crisp research question with success criteria	4
Impact: who benefits, what gap is filled, expected implications	4
Problem Formulation + Related Work (12)	
Coverage of at least 3 closely related lines of work	4
Positioning: comparison table/paragraph clarifying contributions	4
Description of problem with inputs/outputs and formulation	4
Method/Approach (24)	
Description of model/algorithm + training details	6
Justification for design decisions	6
Evaluation strategy (datasets/benchmarks, metrics, baselines)	6
Reproducibility details (framework, hardware, hyperparameters)	6
Data (8)	
Description & datasheet aspects (source, size, balance, biases)	4
Description of preprocessing/augment steps + why and data splits	4
Experiments and Results (24)	
Strong, justified baselines (1–3)	8
Ablations & experiments	8
Visualizations/plots, analysis, and failure modes	8
Conclusion (5)	
Discussion of results and limitations	3
Concrete future work informed by findings	2
Additional (12)	
Clarity, formatting, template compliance (6–8 pages main text)	4
Team contribution table + descriptions of work	4
Proper references cited in consistent bibliographic style	4
Total	100